

AXIOM Medical Research Agent

Orchestrator for the medical research instrument. Composes a sealed medical AXM container with six layer delegates, fires the active profile per source, wraps the per-source bundle in a `MedicalCoordinatorToken`, and renders a bracketed descriptor for downstream LLM use.

The implementation lives in `axiom_medical_agent.py:1`. It is a sibling of `ExoskeletonAgent` — not a subclass — because the workflow differs.

Composition

Component	Role
<code>axiom_medical_container.build_medical_container</code>	Container + core dict
<code>examples/medical_pack.MEDICAL_DELEGATES</code>	6 layer delegates
<code>axiom_event_token.Coordinator</code>	Per-layer compose
<code>axiom_medical_coordinator.MedicalCoordinatorToken</code>	Cross-layer bind
<code>axiom_medical_governance.MedicalGovernanceCheck</code>	Deterministic governance verdict
<code>axiom_medical_descriptor.render</code>	Bracketed LLM-ready descriptor
<code>axiom_medical_ledger</code>	Best-effort signed audit

Layer activation profiles

`LAYER_ACTIVATION_PROFILES` selects which delegates fire for the current question. Call `agent.list_profiles()` for the live list. Typical profiles:

Profile	Use case
<code>summarize</code>	Default — narrative summary of cited sources
<code>mechanism</code>	Pathway / mechanism extraction
<code>evidence</code>	Tier-classified evidence audit
<code>safety</code>	Surface contradictions and safety flags

API

```
from axiom_medical_agent import MedicalResearchAgent

agent = MedicalResearchAgent.from_default_pack(backend=...)
```

```

result = agent.research(
    research_question="What mechanisms link GLP-1 drugs to "
                    "reduced inflammation?",
    sources=[
        {"name": "Cochrane 2023 systematic review",
         "source_type": "systematic_review",
         "text": "..."}
    ],
    profile="mechanism",
)
print(result.descriptor)
print(result.coordinator_tokens[0].to_json(indent=2))

```

`ResearchResult` carries:

- `event_tokens` — per-layer signed `EventToken`s
- `coordinator_tokens` — one `MedicalCoordinatorToken` per source
- `descriptor` — bracketed string for LLM prompt injection
- `manifest_root` — Merkle root over the event tokens
- `requires_human_review` — True if any governance verdict flagged it
- `tier_distribution` — count per evidence tier (1–5)
- `container_id` — sealed container identifier

Constructors

- `MedicalResearchAgent.from_default_pack(backend=...)` — bundled default container
- `MedicalResearchAgent.from_path(axm_path, backend=..., ledger=...)` — load a sealed `.axm`

CANNOT_MUTATE enforcement

`research()` calls `_enforce_cannot_mutate()` before firing. If any field marked `CANNOT_MUTATE` in the sealed container has been silently changed since seal, the call raises `MedicalAgentError` and writes nothing to the ledger.

Evidence tiers

Each source's tier is taken from `evidence_tier` / `source_tier` on the delegate payload; if missing, `classify_source(payload)` is called against `source_type`. Tiers are 1 (highest, e.g. Cochrane systematic review) through 5 (lowest, e.g. expert opinion).

CLI

```
python axiom_medical_agent.py --question "..." --profile mechanism
```

Run with `--profile` to switch activation profiles and `--sources` to point at a JSONL of source dicts.